

Using device: cuda

100% |██████████| 26.4M/26.4M [00:02<00:00, 11.5MB/s]

100% |██████████| 29.5k/29.5k [00:00<00:00, 170kB/s]

100% |██████████| 4.42M/4.42M [00:01<00:00, 3.14MB/s]

100% |██████████| 5.15k/5.15k [00:00<00:00, 24.6MB/s]

[AE] Epoch 1/10: 100% ██████████ 469/469 [00:11<00:00, 54.22it/s]

[AE] Epoch 1, loss=0.0309

[AE] Epoch 2/10: 100% ██████████ 469/469 [00:08<00:00, 44.21it/s]

[AE] Epoch 2, loss=0.0157

[AE] Epoch 3/10: 100% ██████████ 469/469 [00:08<00:00, 55.28it/s]

[AE] Epoch 3, loss=0.0138

[AE] Epoch 4/10: 100% ██████████ 469/469 [00:09<00:00, 54.24it/s]

[AE] Epoch 4, loss=0.0130

[AE] Epoch 5/10: 100% ██████████ 469/469 [00:08<00:00, 39.18it/s]

[AE] Epoch 5, loss=0.0125

[AE] Epoch 6/10: 100% ██████████ 469/469 [00:08<00:00, 55.20it/s]

[AE] Epoch 6, loss=0.0121

[AE] Epoch 7/10: 100% ██████████ 469/469 [00:09<00:00, 59.14it/s]

[AE] Epoch 7, loss=0.0118

[AE] Epoch 8/10: 100% ██████████ 469/469 [00:09<00:00, 56.76it/s]

[AE] Epoch 8, loss=0.0116

[AE] Epoch 9/10: 100% ██████████ 469/469 [00:08<00:00, 51.86it/s]

[AE] Epoch 9, loss=0.0114

[AE] Epoch 10/10: 100% 469/469 [00:08<00:00, 58.31it/s]

[AE] Epoch 10, loss=0.0112

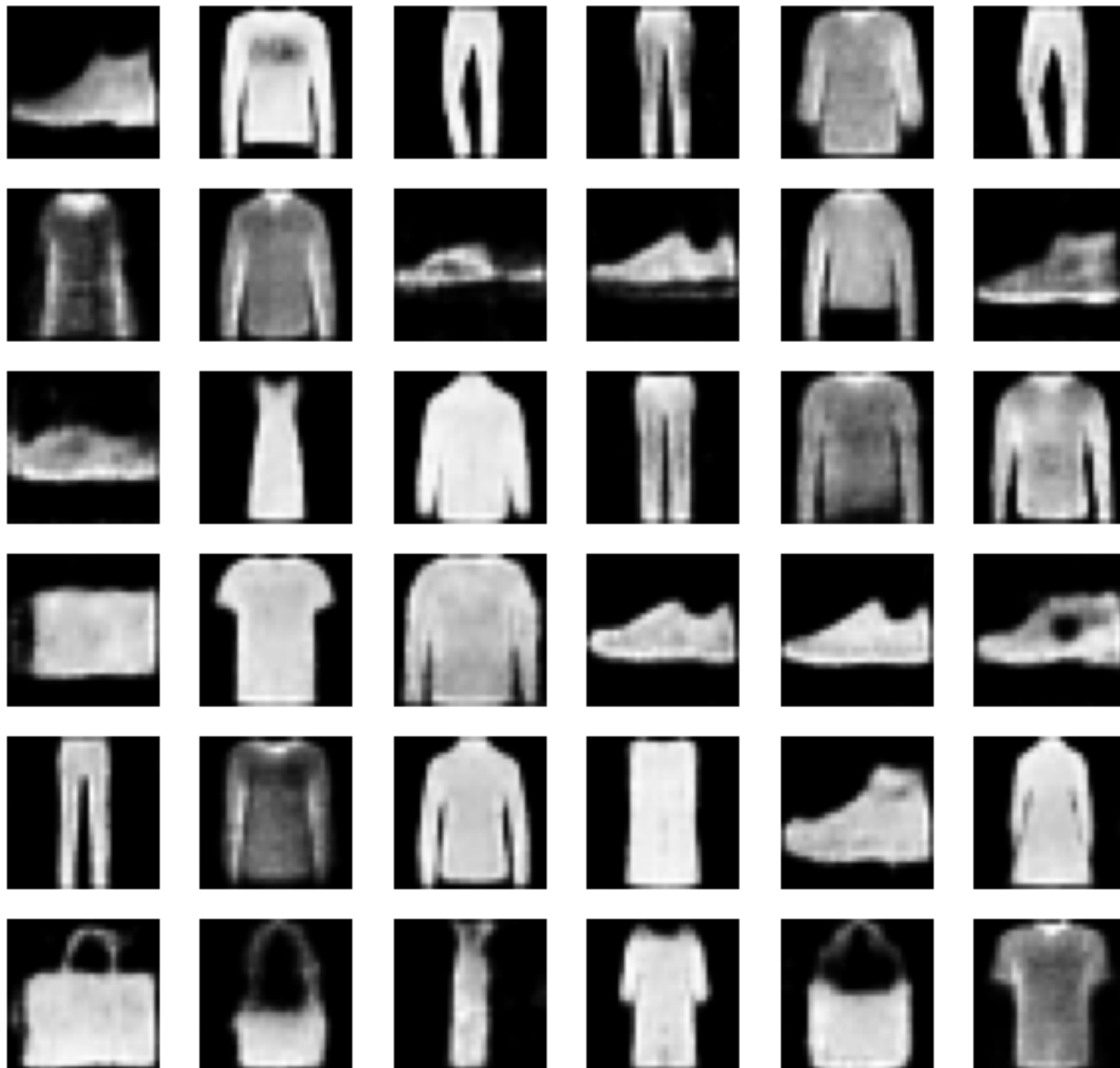
AE - Clean (Figure 1a)



AE - Noisy (Figure 1b)



AE - Denoised (Figure 1c)



[VAE] Epoch 1/12: 100%  469/469 [00:11<00:00, 46.60it/s]

[VAE] Epoch 1, loss=277.1204

[VAE] Epoch 2/12: 100%  469/469 [00:10<00:00, 48.33it/s]

[VAE] Epoch 2, loss=251.7084

[VAE] Epoch 3/12: 100%  469/469 [00:11<00:00, 48.03it/s]

[VAE] Epoch 3, loss=248.1188

[VAE] Epoch 4/12: 100%  469/469 [00:10<00:00, 36.89it/s]

[VAE] Epoch 4, loss=246.3879

[VAE] Epoch 5/12: 100%  469/469 [00:12<00:00, 38.43it/s]

[VAE] Epoch 5, loss=245.1764

[VAE] Epoch 6/12: 100%  469/469 [00:11<00:00, 47.66it/s]

[VAE] Epoch 6, loss=244.4325

[VAE] Epoch 7/12: 100%  469/469 [00:10<00:00, 47.22it/s]

[VAE] Epoch 7, loss=243.8168

[VAE] Epoch 8/12: 100%  469/469 [00:11<00:00, 44.37it/s]

[VAE] Epoch 8, loss=243.3104

[VAE] Epoch 9/12: 100%  469/469 [00:11<00:00, 46.47it/s]

[VAE] Epoch 9, loss=242.8749

[VAE] Epoch 10/12: 100%  469/469 [00:10<00:00, 40.42it/s]

[VAE] Epoch 10, loss=242.5793

[VAE] Epoch 11/12: 100%  469/469 [00:11<00:00, 48.28it/s]

[VAE] Epoch 11, loss=242.2650

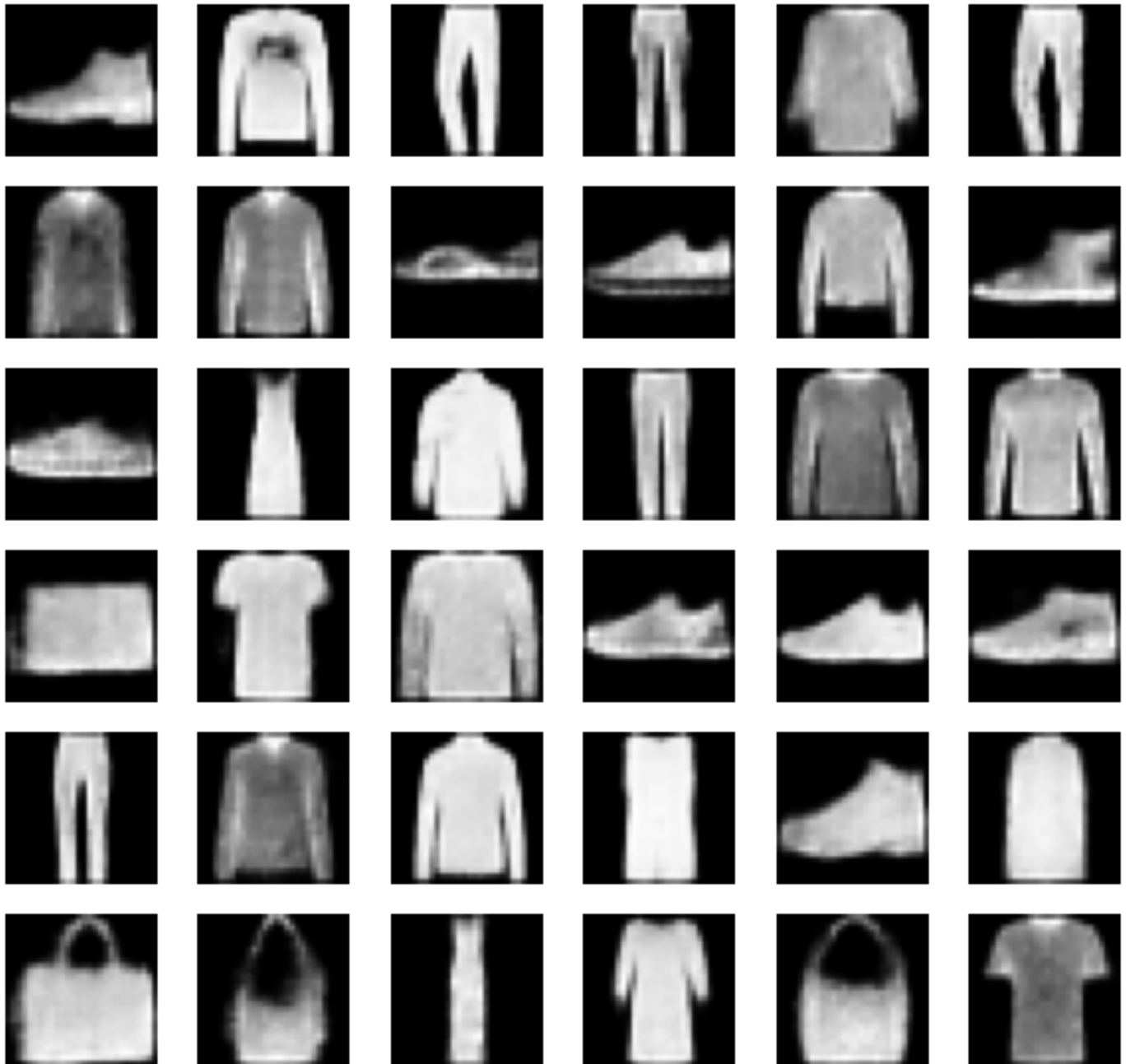
[VAE] Epoch 12/12: 100%  469/469 [00:11<00:00, 45.92it/s]

[VAE] Epoch 12, loss=242.0106

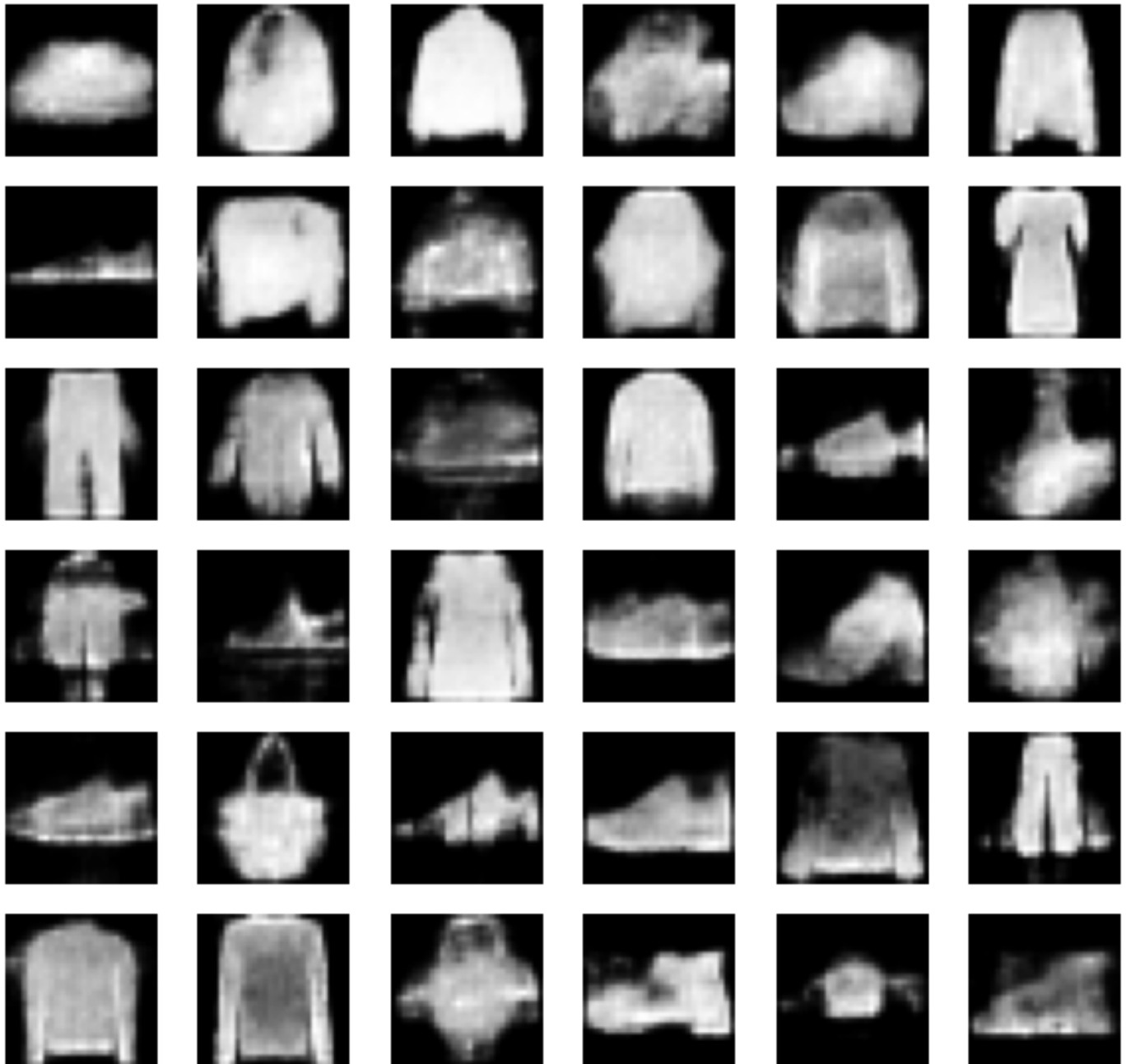
VAE - Original (Recon)



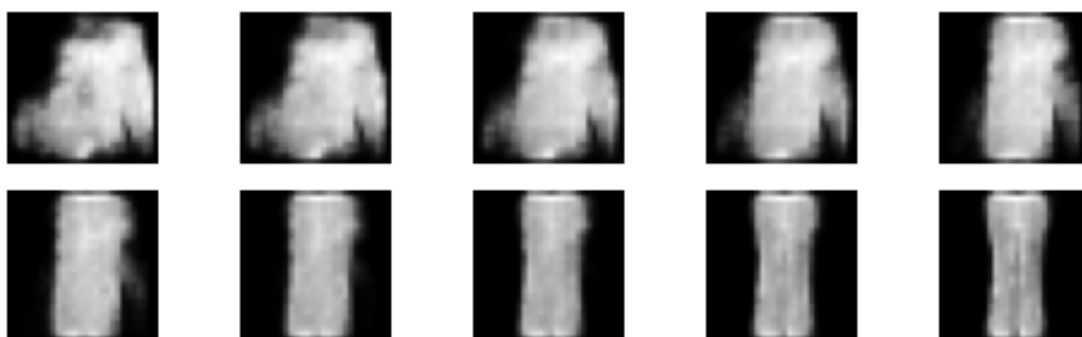
VAE - Reconstruction (Figure 2a)



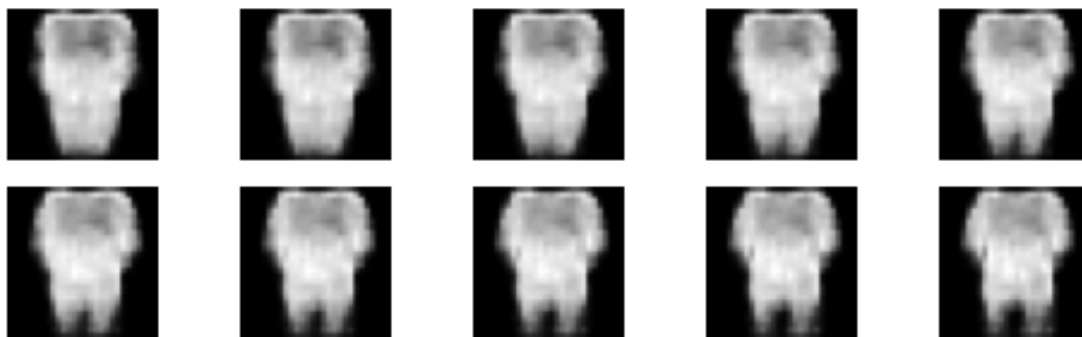
VAE - Random Samples (Figure 2b)



VAE - Interpolations (Figure 3)



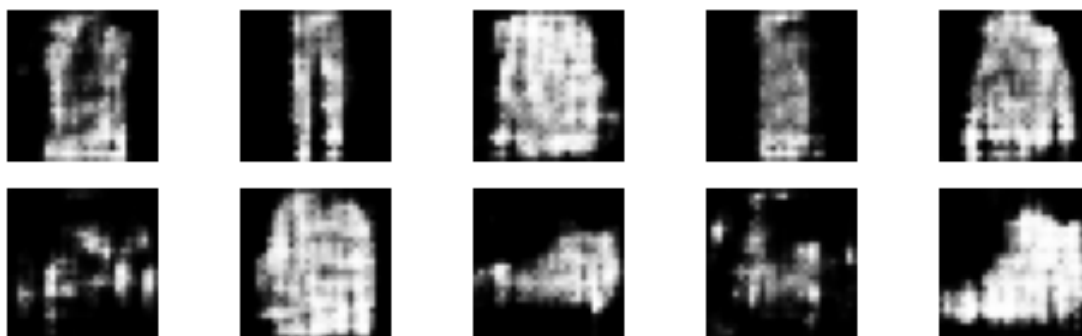
VAE - Traversal dim0 (Figure 4)



[cGAN] Epoch 1/12: 100%  469/469 [00:19<00:00, 21.62it/s]

[cGAN] Epoch 1, D_loss=1.2137, G_loss=1.3112

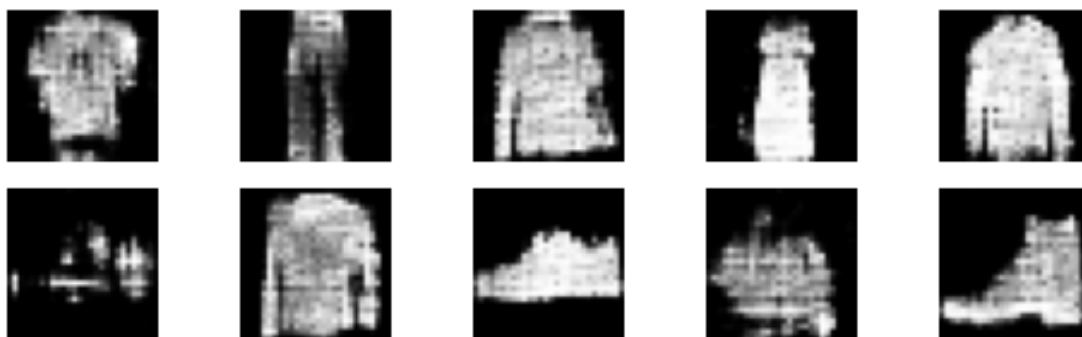
cGAN - Class Samples (Figure 5, epoch 1)



[cGAN] Epoch 2/12: 100%  469/469 [00:19<00:00, 23.08it/s]

[cGAN] Epoch 2, D_loss=1.2206, G_loss=0.9020

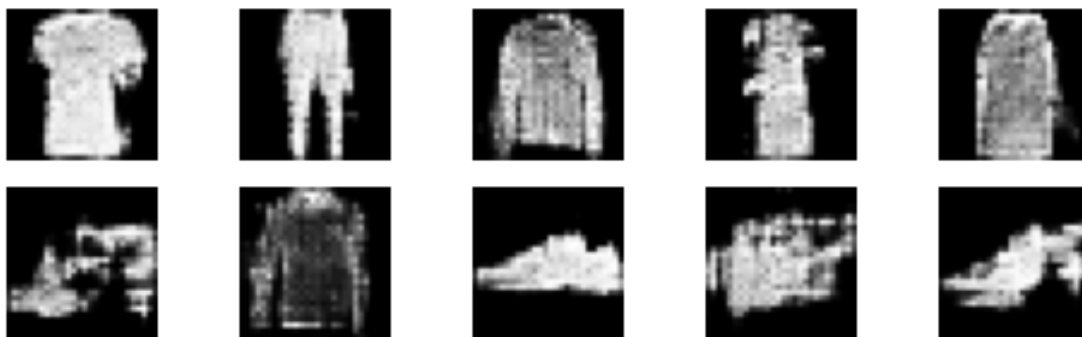
cGAN - Class Samples (Figure 5, epoch 2)



[cGAN] Epoch 3/12: 100%  469/469 [00:19<00:00, 24.54it/s]

[cGAN] Epoch 3, D_loss=1.2481, G_loss=0.9780

cGAN - Class Samples (Figure 5, epoch 3)



[cGAN] Epoch 4/12: 100%  469/469 [00:18<00:00, 25.60it/s]

[cGAN] Epoch 4, D_loss=1.3488, G_loss=0.8429

cGAN - Class Samples (Figure 5, epoch 4)



[cGAN] Epoch 5/12: 100%  469/469 [00:18<00:00, 24.81it/s]

[cGAN] Epoch 5, D_loss=1.3500, G_loss=0.8222

cGAN - Class Samples (Figure 5, epoch 5)



[cGAN] Epoch 6/12: 100%  469/469 [00:19<00:00, 22.98it/s]

[cGAN] Epoch 6, D_loss=1.3223, G_loss=0.8665

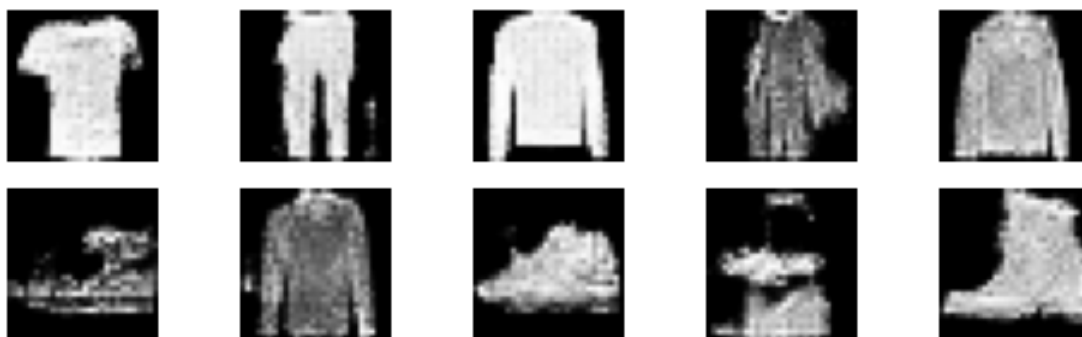
cGAN - Class Samples (Figure 5, epoch 6)



[cGAN] Epoch 7/12: 100%  469/469 [00:19<00:00, 25.13it/s]

[cGAN] Epoch 7, D_loss=1.3612, G_loss=1.0070

cGAN - Class Samples (Figure 5, epoch 7)



[cGAN] Epoch 8/12: 100%  469/469 [00:19<00:00, 25.14it/s]

[cGAN] Epoch 8, D_loss=1.3743, G_loss=0.8491

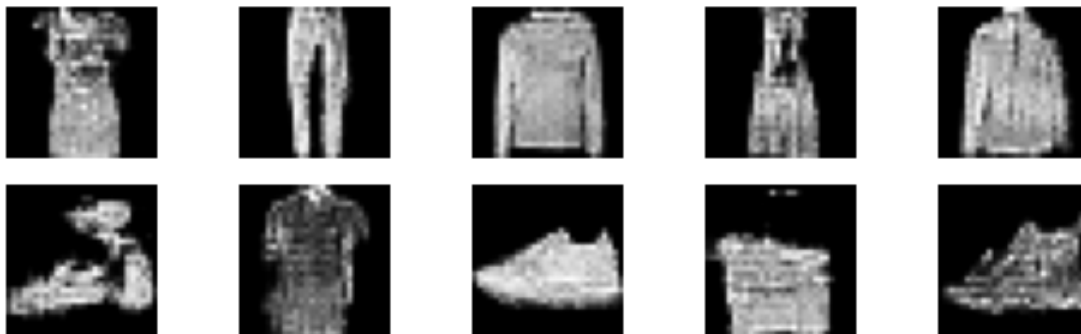
cGAN - Class Samples (Figure 5, epoch 8)



[cGAN] Epoch 9/12: 100%  469/469 [00:19<00:00, 24.14it/s]

[cGAN] Epoch 9, D_loss=1.3753, G_loss=0.9006

cGAN - Class Samples (Figure 5, epoch 9)



[cGAN] Epoch 10/12: 100%  469/469 [00:19<00:00, 24.84it/s]

[cGAN] Epoch 10, D_loss=1.3405, G_loss=0.8679

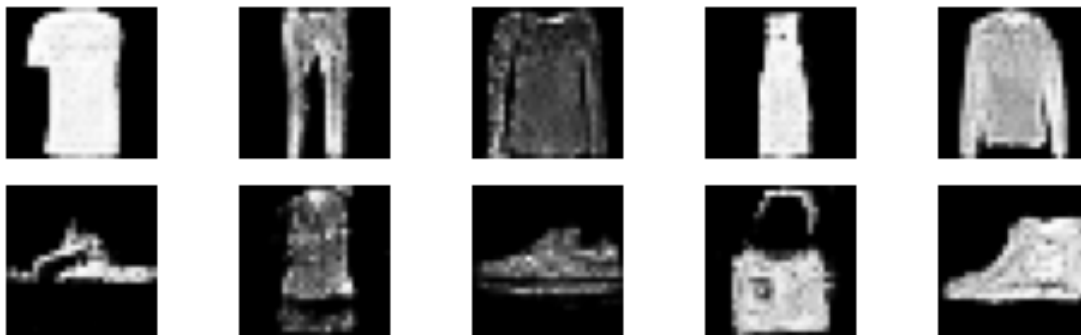
cGAN - Class Samples (Figure 5, epoch 10)



[cGAN] Epoch 11/12: 100%  469/469 [00:19<00:00, 24.87it/s]

[cGAN] Epoch 11, D_loss=1.3381, G_loss=0.9196

cGAN - Class Samples (Figure 5, epoch 11)



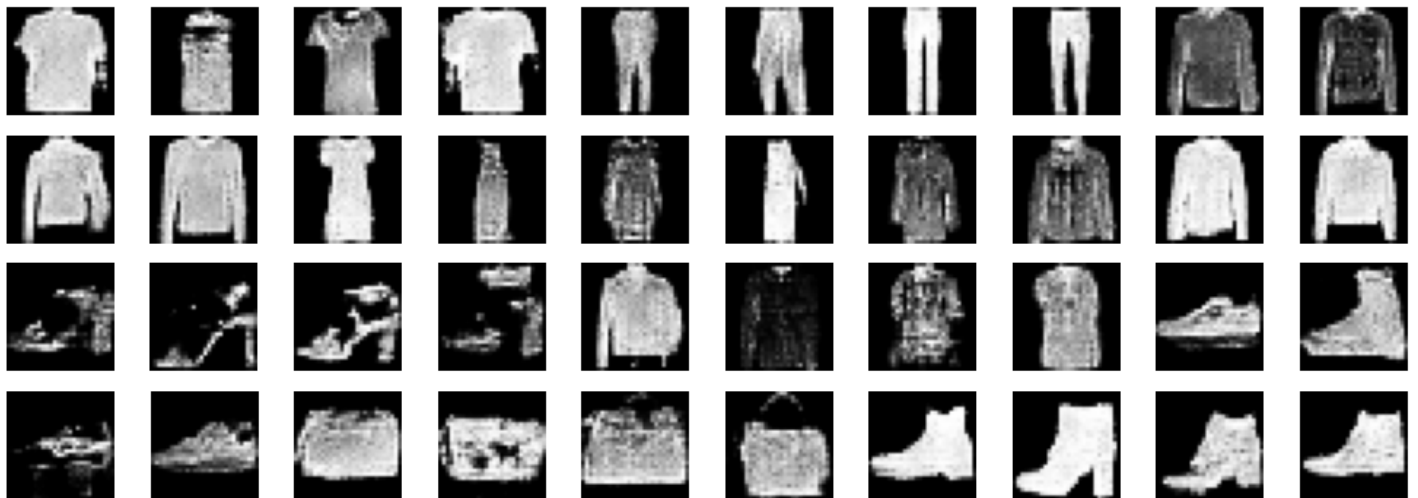
[cGAN] Epoch 12/12: 100%  469/469 [00:20<00:00, 21.18it/s]

[cGAN] Epoch 12, D_loss=1.3942, G_loss=0.8847

cGAN - Class Samples (Figure 5, epoch 12)



cGAN - Diversity Grid (Figure 6)



AdaIN - Content (Figure 7a)



AdaIN - Style (Figure 7b)



AdaIN - Mixed (Figure 7c)

